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337313 (37)

BE (3rd Semester)

Examination, Nov.-Dec., 2014

Branch : Mechanical

MATERIAL SCIENCE

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Answer all five questions. Part (a) of each question is compulsory. attempt any two parts from (b), (c) and (d).

Q. 1. (a) Define unit cell. 2

(b) Discuss surface defects with neat sketches. 7

(c) What are dendrites ? Explain the mechanism of solidification of metals. 7

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(d) What is grain ? Describe the effect of grain size and grain boundary on properties of materials. 7

Q. 2. (a) Define resilience. 2

(b) Explain yield point phenomenon and related effects. 7

(c) Define Bauschinger effect. Explain utility of this effect in case of cold working of metals. 7

(d) What is the purpose of performing tensile test for a metal ? Write the principle and procedure of determination of tensile test of a metal 7

Q. 3. (a) Define Eutectic system 2

(b) Draw and explain phase diagram of Iron-Carbon system

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(c) Construct and explain equilibrium diagram of two metals A and B when component A and B have liquid solubility with partial solid solubility. 7

(d) Draw cooling curve for pure iron and explain the curve. 7

Q. 4. (a) Define hardenability. 2

(b) Sketch and explain TTT curve for a specific composition of steel. 7

(c) Differentiate between austempering and martempering. 7

(d) What are the advantages of hardening the surfaces only ? Describe nitriding as a method of surface hardening of steel. 7

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Q. 5. (a) Give composition of wild steel. 2

(b) Write short note on : 7

(i) Nodular cast iron

(ii) Gun metal

(c) Give composition, properties and uses of the following : 7

(i) Naval brass

(ii) Sintered carbide

(d) Define alloys. Write industrial applications of light metal alloys. 7

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